

Shivang Chopra

Ph.D. in Computer Science at Georgia Institute of Technology

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RESEARCH INTERESTS

Robust Representation Learning: Investigating the algorithmic and geometric principles underlying generalization and efficiency in multimodal foundation models, with a focus on transformer architecture design, and representation structure.

Adaptive Computation and Efficient Inference: Developing methods to scale virtual depth and adaptive computation at inference time, enabling pretrained LLMs to achieve deeper reasoning capabilities without increasing total parameter counts.

Data-Efficient Learning and Training Optimization: Designing data selection, active learning, and manifold-aware training methods to improve sample efficiency, training stability, and generalization in large-scale multimodal and language models.

EDUCATION

Georgia Institute of Technology

Jan 2025 - May 2029 (expected)

Ph.D. in Computer Science

Advisor: Prof. Zsolt Kira [🔗](#)

Publications: CVPR 2026 [2], CVPR(W) 2025 [8], CVPR 2025 [5]

Georgia Institute of Technology

Aug 2022 - Dec 2024

Master of Science in Computer Science

CGPA: 3.9 / 4.0

Thesis: CRAFT: Curriculum Rank Adversarial Fine-Tuning for Robust Vision Language Models [📄](#)

Publications: WACV 2025 [3], CoRL 2023 [6], ICML(W) 2022 [10]

Delhi Technological University

Aug 2016 - May 2020

Bachelor of Technology in Computer Engineering

CGPA: 8.6 / 10.0

Publications: AAAI 2020 [4], IEEE CDS [8]

RESEARCH EXPERIENCE

Google Research

California, USA

Student Researcher

May 2026 - Aug 2026

Mentor: Gaurav Menghani [🔗](#)

Sony R&D Labs

Zurich, Switzerland

Computer Vision Research Intern

Sep 2023 - Apr 2024

Mentor: Valentina Cavinato [🔗](#)

Amazon Science

Washington, USA

Applied Scientist Intern

May 2023 - Aug 2023

Mentor: Dr. Tarang Chugh [🔗](#)

Microsoft Research Lab India

Bangalore, India

Research Intern

May 2022 - Aug 2022

Mentors: Dr. Venkat Padmanabhan [🔗](#), Dr. Saumay Pushp [🔗](#)

IBM Research Lab

Delhi, India

Research Intern

June 2020 - Aug 2020

Mentor: Dr. Sameep Mehta [🔗](#), Nistha Madaan [🔗](#)

PUBLICATIONS

Under Review

- [1] **Trajectory Aligned Gradient Suppression for Test-Time Generalization in Recurrent Transformers**
Shivang Chopra, Zhang Xi-Jia, Karmesh Yadav, Zsolt Kira.

Conferences and Journals

- [2] **The Geometry of Robustness: Optimizing Loss Landscape Curvature and Feature Manifold Alignment for Robust Finetuning of Vision-Language Models**

Shivang Chopra, Shaunak Halbe, Chengyue Huang, Brisa Maneechotesuwan, Zsolt Kira.

Paper accepted at IEEE/CVF Conference on computer vision and pattern recognition, [CVPR 2026](#) [📄](#)

- [3] **Refining Text-to-Image Generation: Towards Accurate Training-Free Glyph-Enhanced Image Generation**

Sanyam Lakhanpal, Shivang Chopra, Vinija Jain, Aman Chadha, Man Luo

Published at IEEE/CVF Winter Conference on Applications of Computer Vision, [WACV 2025](#) [📄](#)

- [4] **Hindi-English Hate Speech Detection: Author Profiling, Debiasing, and Practical Perspectives**

Shivang Chopra, Ramit Sawhney, Puneet Mathur, Rajiv Ratn Shah

Published at the AAAI Conference on Artificial Intelligence, [AAAI 2020](#) [📄](#)

[5] **FRAMES-VQA: Benchmarking Fine-Tuning Robustness across Multi-Modal Shifts in Visual Question Answering**
Chengyue Huang*, Brisa Maneechotesuwan*, **Shivang Chopra**, Zsolt Kira
Published at IEEE/CVF Conference on Computer Vision and Pattern Recognition, [CVPR 2025](#)

[6] **Learning to Discern: Imitating Heterogeneous Human Demonstrations with Preference and Representation Learning**
Sachit Kuhar, Shuo Cheng, **Shivang Chopra**, Matthew Bronars, Danfei Xu
Published at the Conference on Robot Learning, [CoRL 2023](#)

[7] **Directional Gradient Projection for Robust Fine-tuning of Foundation Models**
Chengyue Huang, Junjiao Tian, Brisa Maneechotesuwan, **Shivang Chopra**, Zsolt Kira
Published at the International Conference on Learning Representations, [ICLR 2025](#)

[8] **Attention-Net: An Ensemble Sketch Recognition Approach using Vector Image**
Shivang Chopra*, Gaurav Jain*, Suransh Chopra*, Anil Singh Parihar
Published in IEEE Transactions on Cognitive and Developmental Systems, [IEEE CDS 2022](#)

Workshops/Preprint

[8] **MedMoE: Modality-Specialized Mixture of Experts for Medical Vision-Language Understanding**
Shivang Chopra, Lingchao Mao, Gabriela Sanchez-Rodriguez, Andrew J. Feola, Jing Li, Zsolt Kira
Published at the [MMFF-BIOMED workshop at CVPR 2025](#).

[9] **Source-free domain adaptation with diffusion-guided source data generation**
Shivang Chopra, Suraj Kothawade, Houda Aynaou, Aman Chadha. [arXiv 2024](#)

[10] **Active Data Discovery: Mining Unknown Data using Submodular Information Measures**
Suraj Kothawade, **Shivang Chopra**, Saikat Ghosh, Rishabh Iyer
Published at the [RealML workshop at ICML 2022](#).

Patent

[P1] Hybrid Environment for Interactions Between Virtual and Physical Users (**US 18/358,485**), 2024

AWARDS AND ACHIEVEMENTS

Graduate Student Scholarship , Georgia Institute of Technology	2023,2024
Research Excellence Award , Delhi Technological University	2020
Student Research Travel Grant , Microsoft Research India	2020
Qualified for Onsite Regionals , ACM ICPC (International Collegiate Programming Contest)	2018

RELEVANT COURSEWORK

Ph.D. / Master Of Science: Machine Learning, Vision Language Models, Deep Learning for Robotics, Game AI, Natural Language Processing, Learning and Memory, Cognitive Psychology

Bachelor of Technology: Machine Learning, Artificial Intelligence, Natural Language Processing, Distributed Systems, Operating Systems, Computer Networks, Information and Network Security, Computer Organization and Architecture

TECHNICAL SKILLS

Programming Languages	Python, Bash, C, C++
Deep Learning Frameworks	PyTorch, TensorFlow, JAX, Accelerate, Pytorch Lightning
Distributed System Tools	Slurm, Docker, Kubernetes

COMMUNITY AND ACADEMIC SERVICES

Reviewer: CVPR 2026, NeurIps 2024-25, ICLR 2025, ICML 2025

Student Volunteer: AAAI 2020, ICRA 2025

Project Lead, AI@GT (Georgia Tech): Mentoring a group of three undergraduate students to build self-evolving embodied agents that learn by interacting with their environment.

Admin, GT Skynet: Part of the team maintaining Georgia Tech’s SLURM-based GPU cluster to support ML research and coursework.